

**Time:** 5 minutes**Outcome:** Use marginal benefits and costs while avoiding sunk-cost reasoning.**Difficulty:** ●○○ Beginner

Marginal Analysis



THE CORE IDEA

Marginal analysis compares the additional benefit of the next unit or small change with its additional cost. A decision should expand while marginal benefit is at least marginal cost and stop before the next unit costs more than it adds. Sunk costs do not affect this forward-looking comparison, while costs or benefits imposed on others matter in social marginal analysis.



HOW TO RECOGNIZE IT

- Look for a decision about one more unit, one more hour, or a small change in an activity.
- Compare the additional benefit of that change with its additional cost.
- Continue while marginal benefit is at least marginal cost; stop before the next unit costs more than it adds.
- Ignore totals or averages when the question asks whether the next unit is worthwhile.



WATCH OUT

A common mistake is to decide from total benefit or total cost. Marginal analysis compares the additional benefit and additional cost of the next unit.



WORKED EXAMPLE: COMPARING MARGINAL BENEFIT AND COST

A warehouse earns \$15 from handling one additional box, and the marginal cost of that box is \$13. Marginal net benefit is $\$15 - \$13 = \$2$, so handling the box increases profit. If the next box cost \$17 instead, its marginal net benefit would be $-\$2$ and the warehouse should stop before that box.



CHECK YOURSELF

Total cost is \$100 for 10 units, and producing an eleventh unit adds \$15. Which number is average cost, and which is marginal cost?

**READY?**

Return to the game and master this concept.